





Diffraction of Cathode Rays by a Thin Film.

IF a fine beam of homogeneous cathode rays is sent nearly normally through a thin celluloid film (of the order 3×10^{-6} cm. thick) and then received on a photographic plate 10 cm. away and parallel to the film, we find that the central spot formed by the undeflected rays is surrounded by rings, recalling in appearance the haloes formed by mist round the sun. A photograph so obtained is reproduced (Fig. 1). If the density of the plate is measured by a photometer at a number of points along a radius, and the intensity of the rays at these points found by using the characteristic blackening curve of the plate (see *Phil. Mag.*, vol. 1, p. 963, 1926), the rings appear as humps on the intensity-distance curves. In this way rings can be

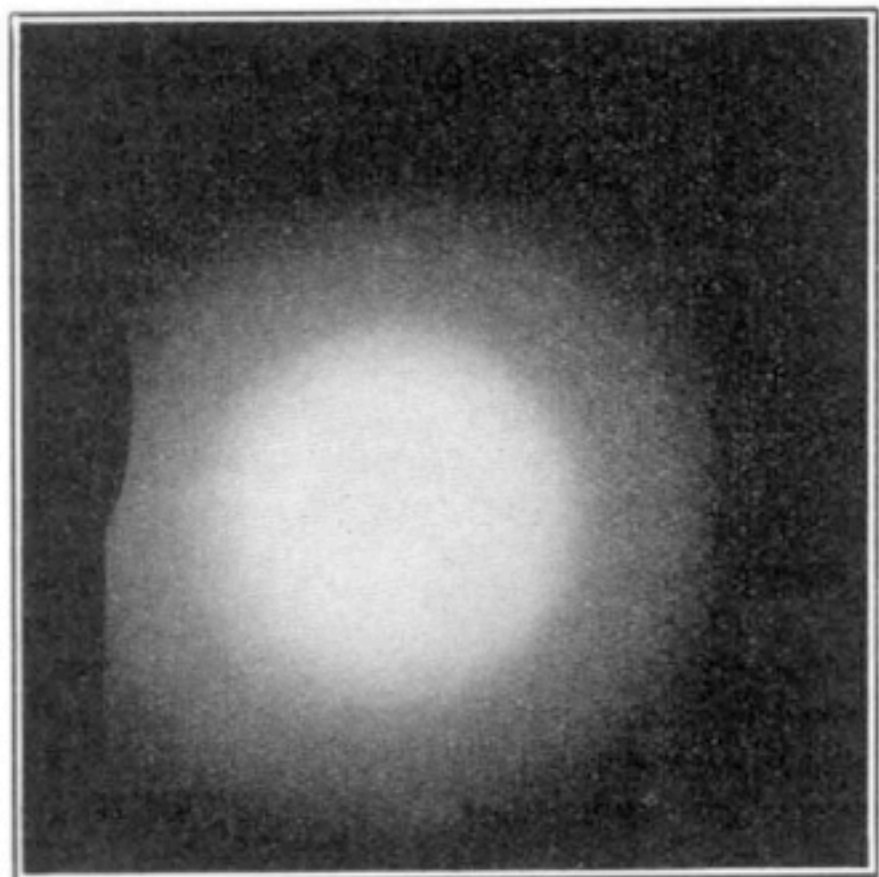
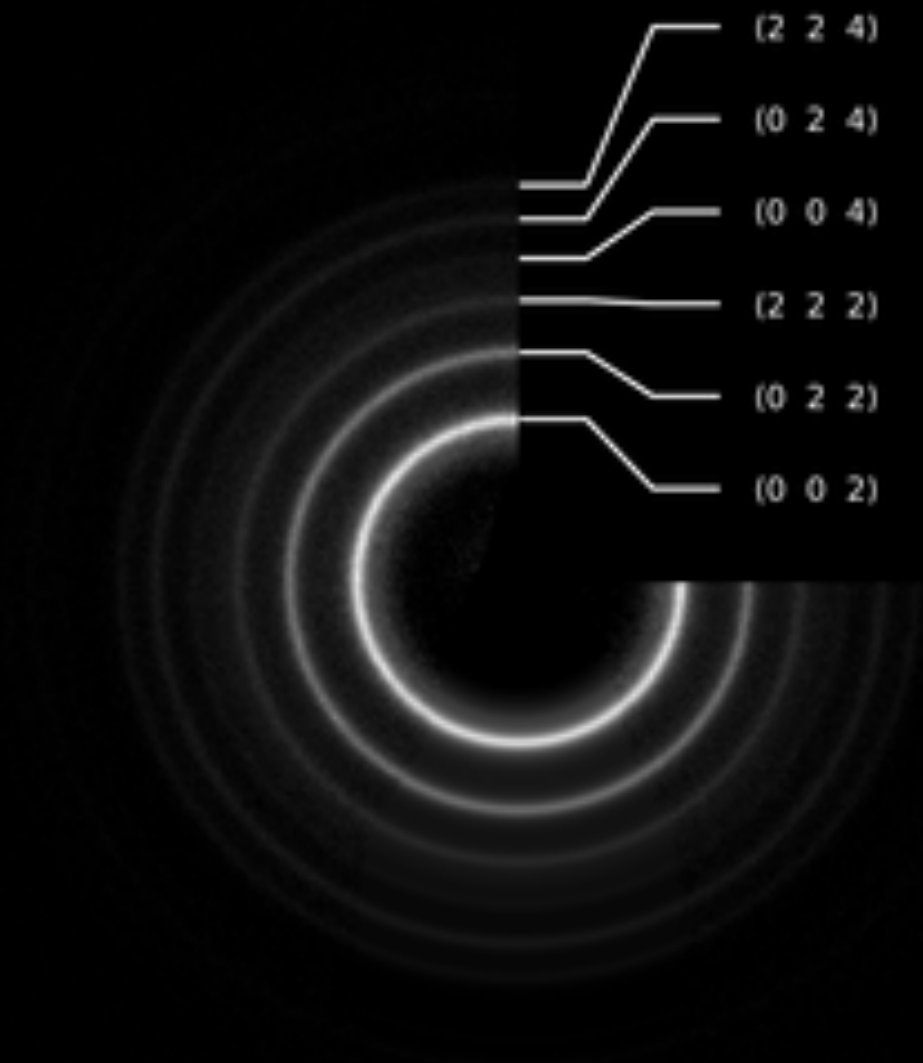
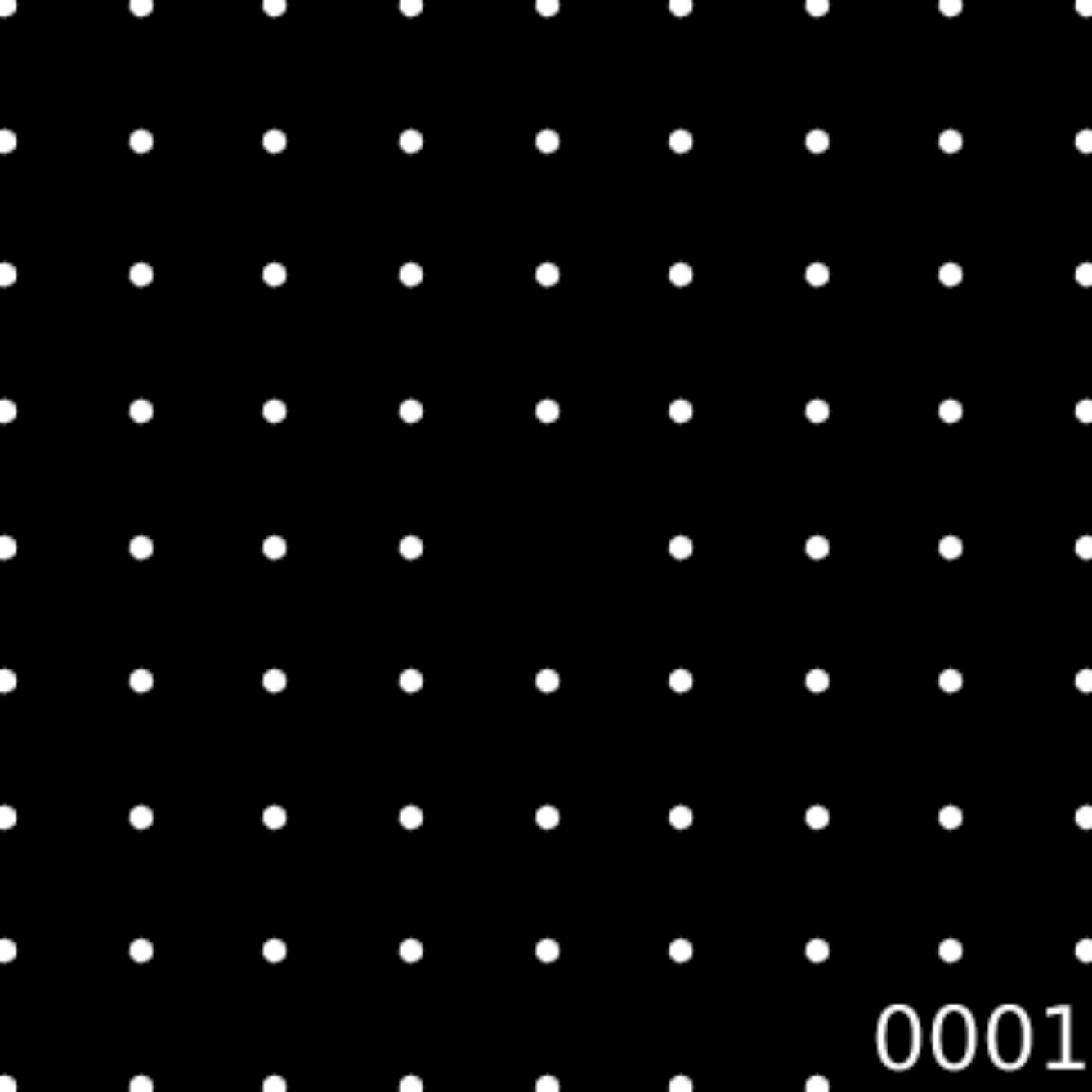


FIG. 1.

detected which may not be obvious to direct inspection. With rays of about 13,000 volts two rings have been found inside the obvious one. Traces have been found of a fourth ring in other photographs, but not more than three have been found on any one exposure. This is probably due to the limited range of intensity within which photometric measurements are feasible.

Nature 119, 890 (1927)





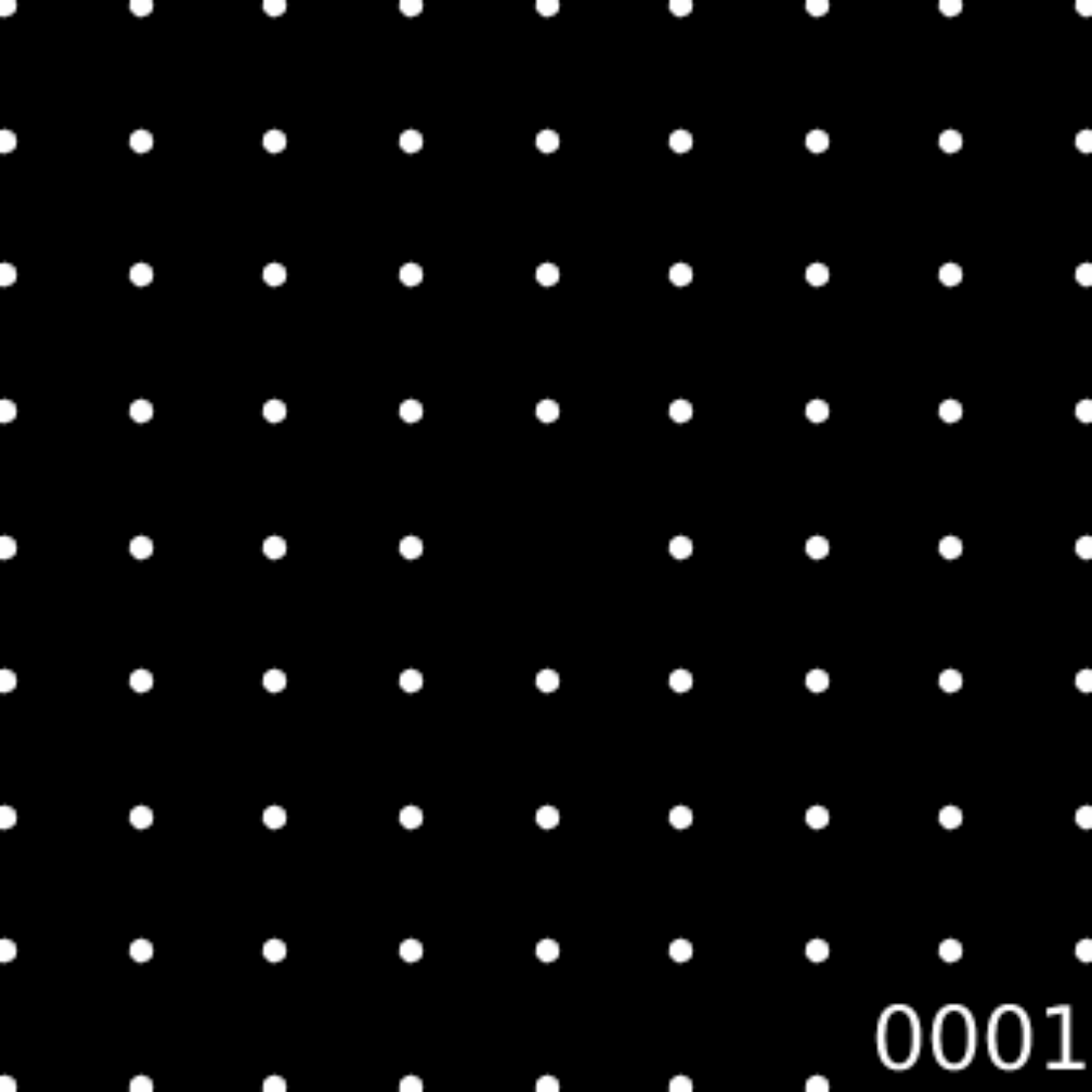
0001

Взаимосвязь между точечной и
кольцевой дифракцией на примере от 1
до 1000 зерен MgO , симуляция в
CrysTBox.

Эксперимент по дифракции электронов на кристалле MgO

Источники риска: ВИКИПЕДИЯ

Диффракция электронов
важна в современных
электронных микроскопах



0001

Электрон это волна

От открытия до наших дней

Диффракция электронов
важна в современных
электронных микроскопах

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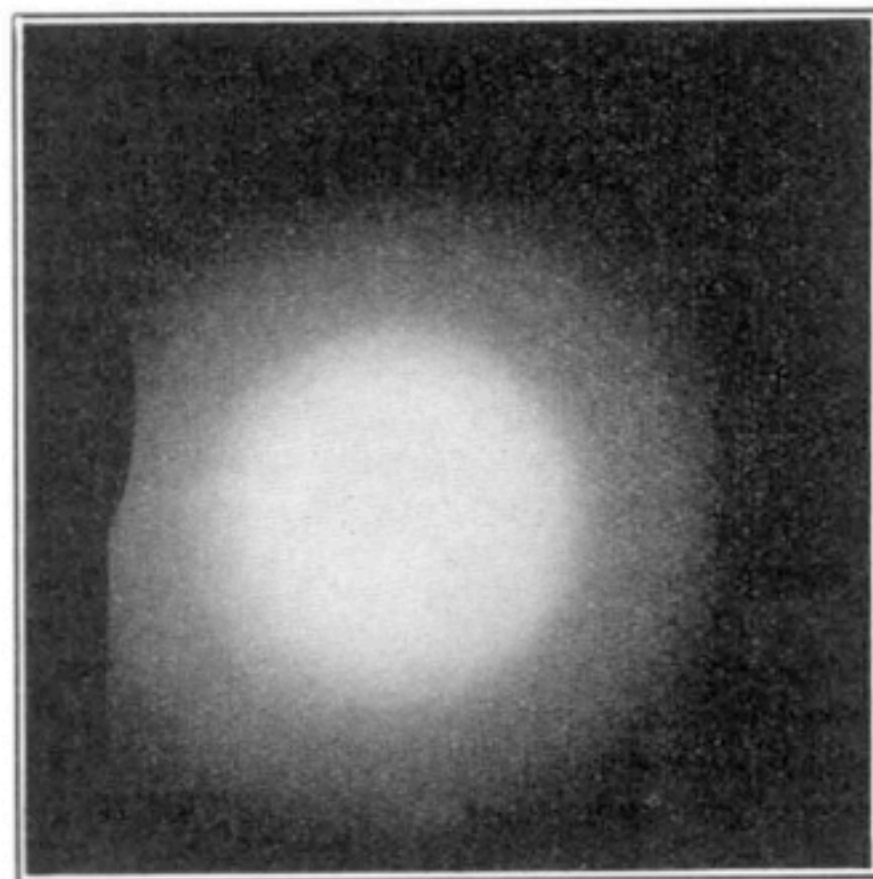
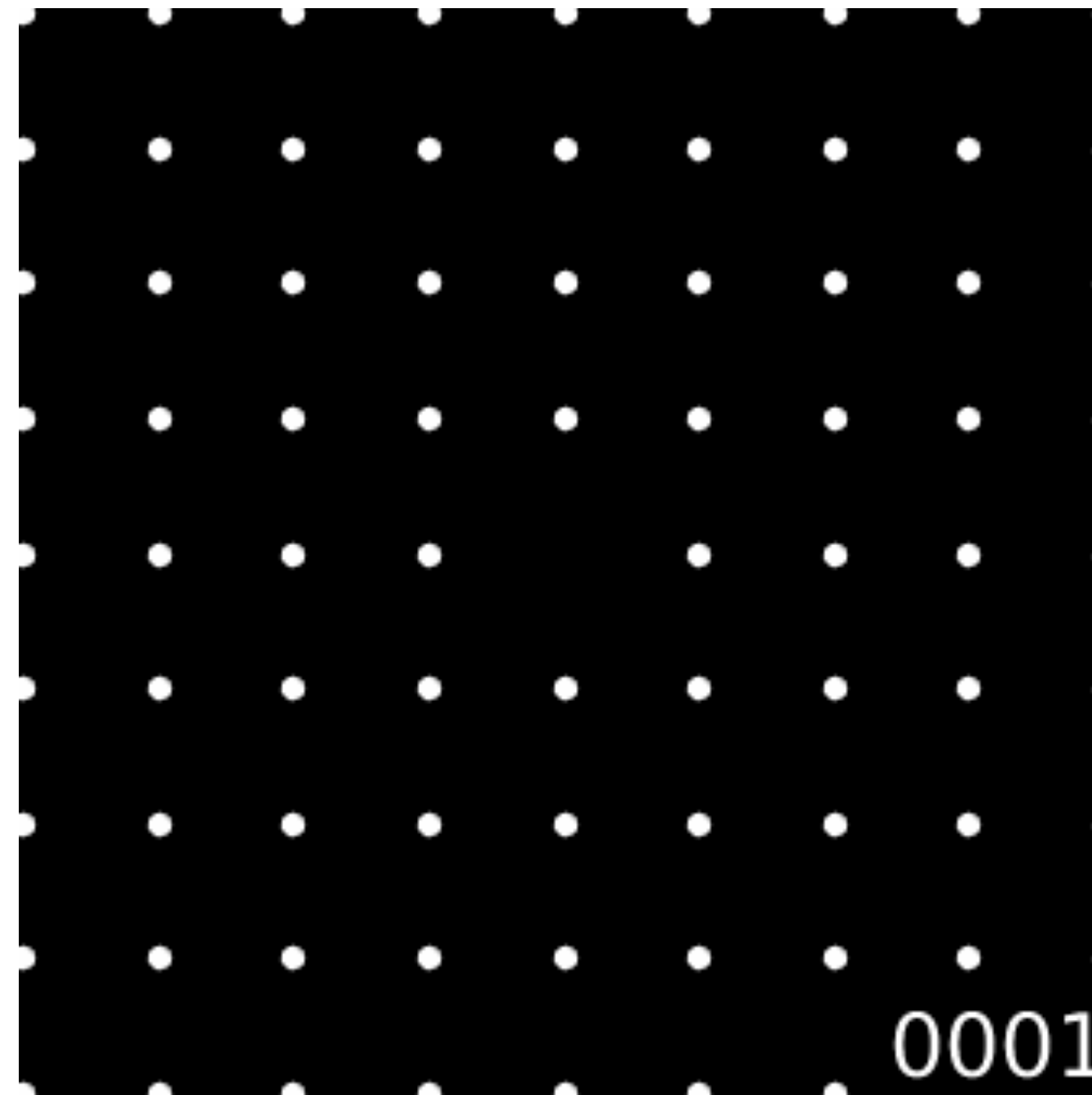


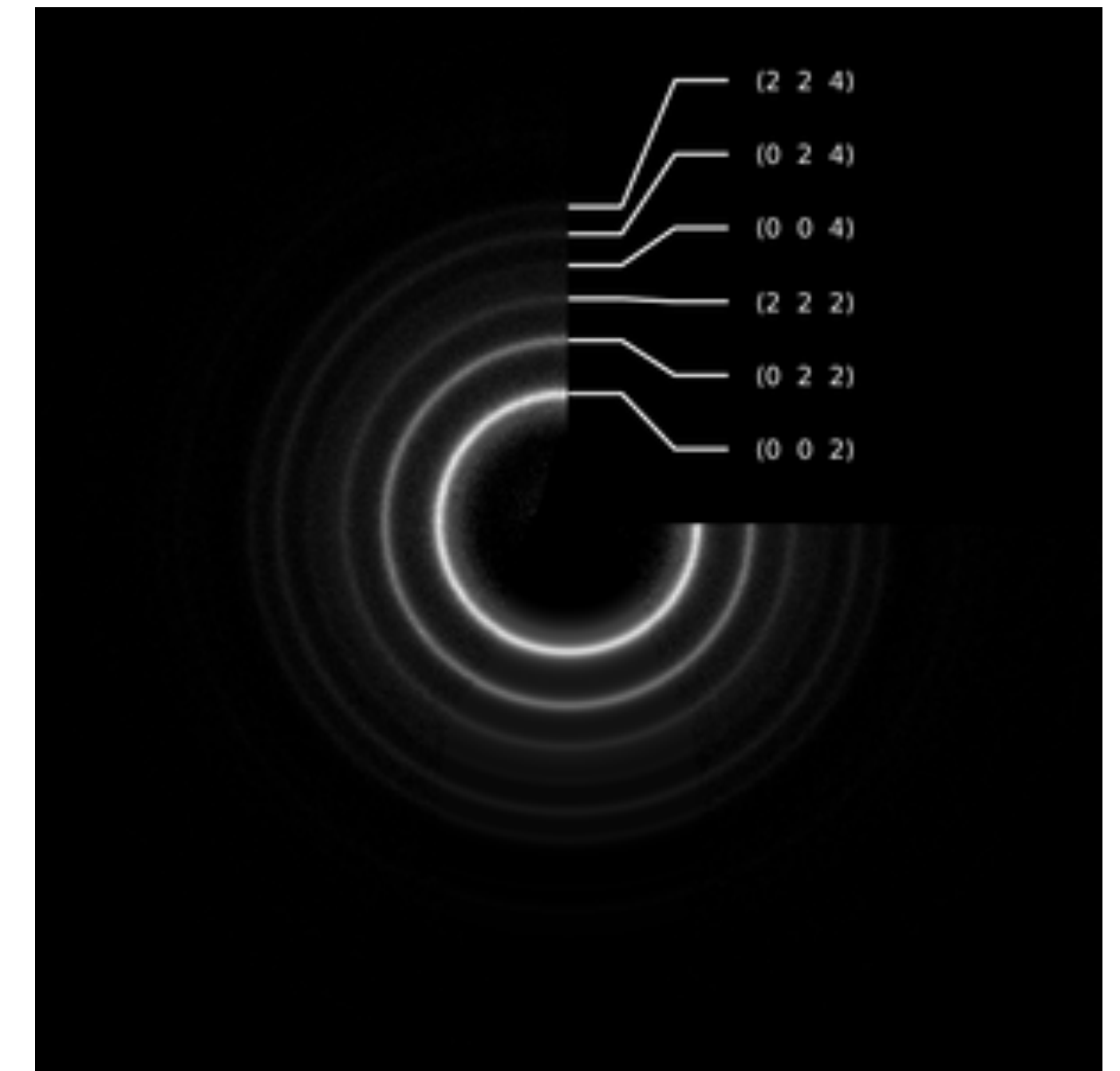
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Эксперимент по дифракции
электронов на кристалле MgO

Источник рисунков на слайде: википедия

Электрон в атоме